



Rural Health Kiosk

AI-Powered Healthcare Access for Underserved Communities



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What the Project Does

An AI-powered rural health kiosk that expands healthcare access in underserved communities, letting patients interact with a virtual assistant via voice and text.

Built on Retrieval-Augmented Generation (RAG), multimodal AI, and remote diagnostics, it supports symptom assessment, triage, and personalized care. As part of the CoDiRA initiative, it emphasizes culturally sensitive, multilingual, and community-driven delivery.

Why This Matters?

- › Rural and underserved populations face limited access to healthcare
- › Shortage of medical professionals in remote areas
- › Delayed diagnosis leads to preventable complications
- › Existing chatbots lack realism and bedside manner

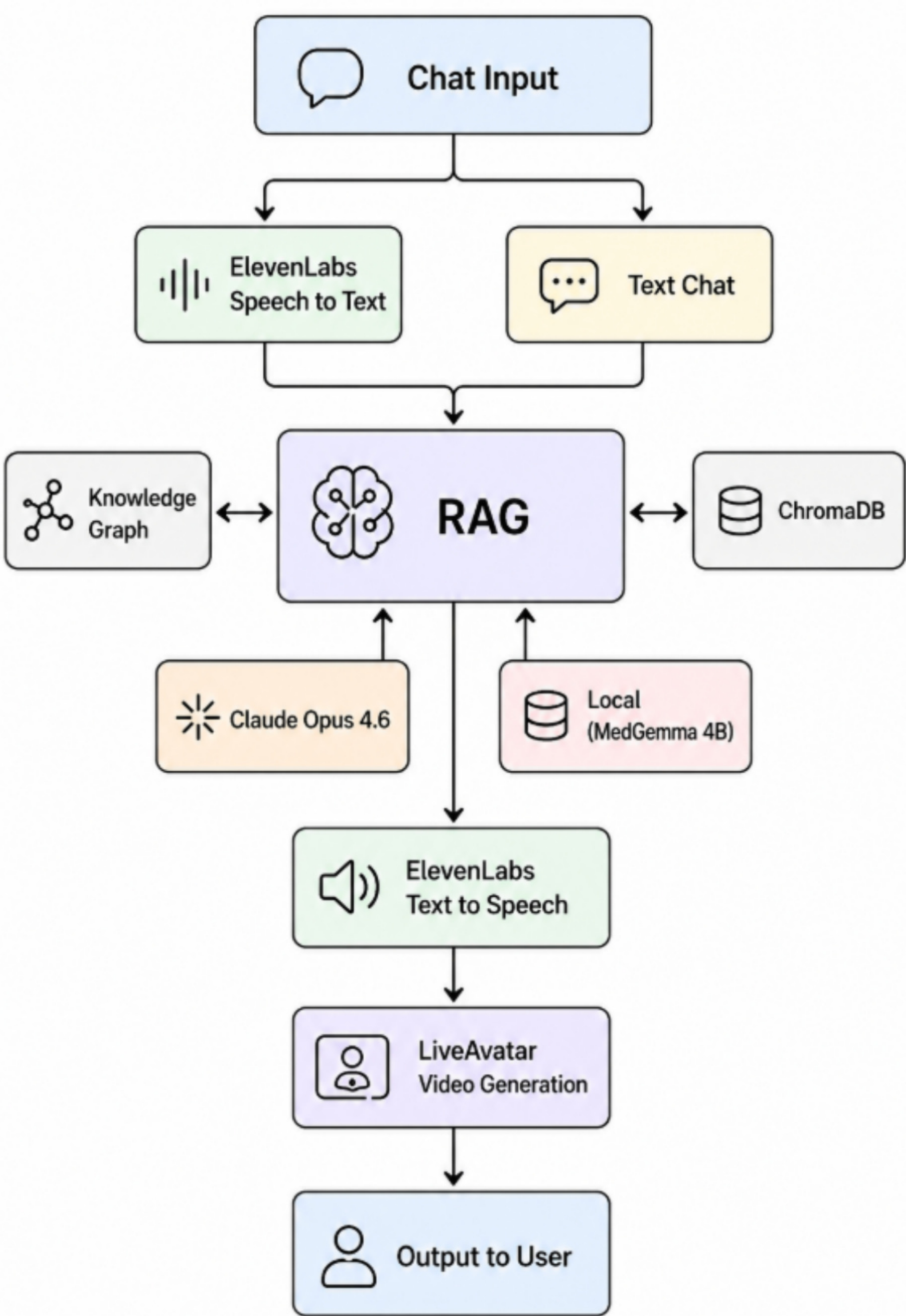
Our Solution

A smart AI-powered healthcare kiosk that provides accessible medical support through voice, text, and empathetic interaction. Designed for underserved populations, it pairs advanced tech with multilingual support, virtual education and community collaboration.

What Makes It Unique?

- › Fine-tuned MedGemma model for bedside manner
- › Real-time avatar interaction
- › Voice-to-voice AI communication
- › Retrieval-Augmented Generation for medical context
- › Designed for training and communication practice

System Workflow



Technology Stack

MedGemma 4B · Claude Opus 4.6 · Whisper · ElevenLabs · HeyGen · FastAPI · Docker · Jetstream GPU

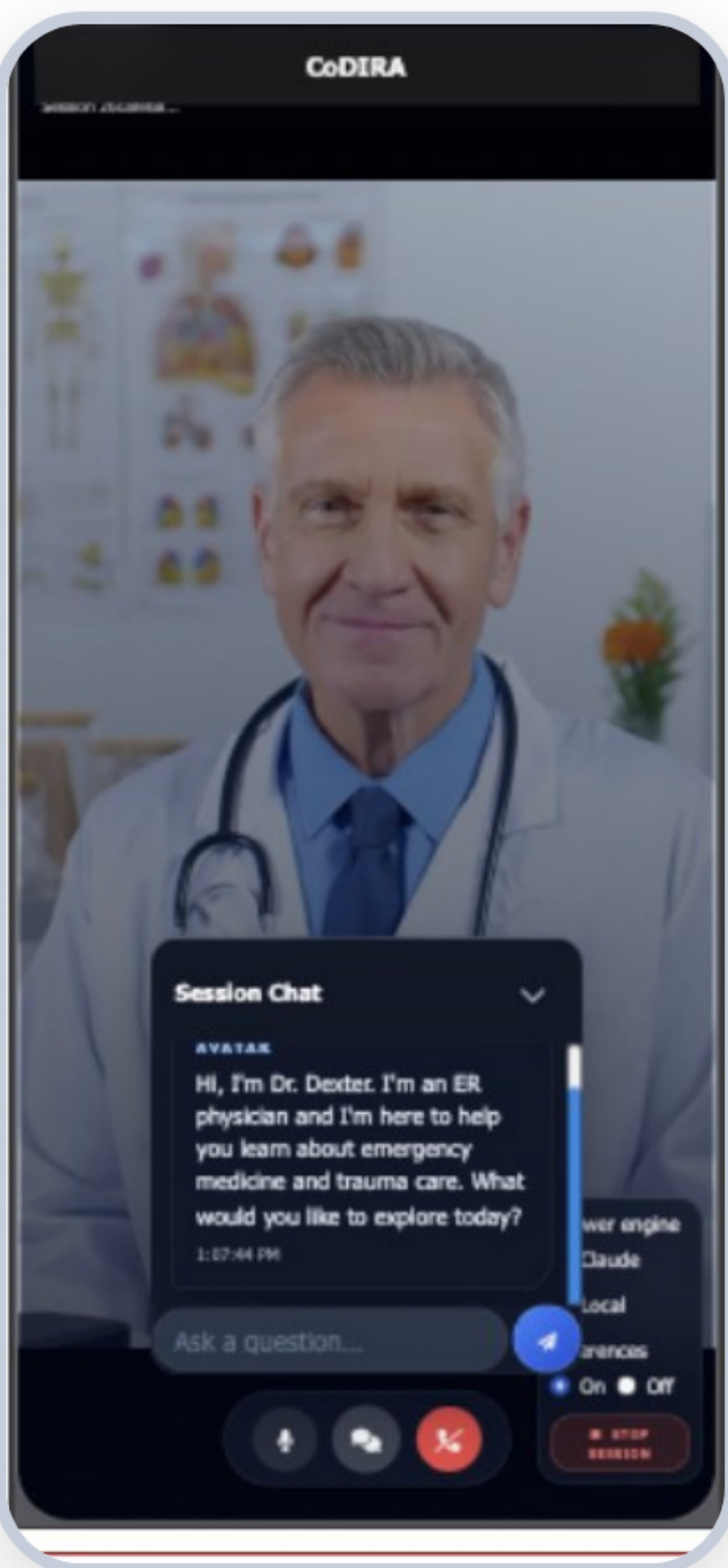
Outcomes

- › Improved empathetic AI responses
- › Early detection through AI-assisted screening
- › Reduced dependency on physical healthcare infrastructure
- › Scalable healthcare model for global deployment

Real-World Applications

- › Rural healthcare kiosks
- › Community health centers
- › Remote patient triage and screening
- › Mobile healthcare units
- › Public health awareness and education

Dr. Dexter



Live Demo



Scan to Experience Dr.Dexter Live